

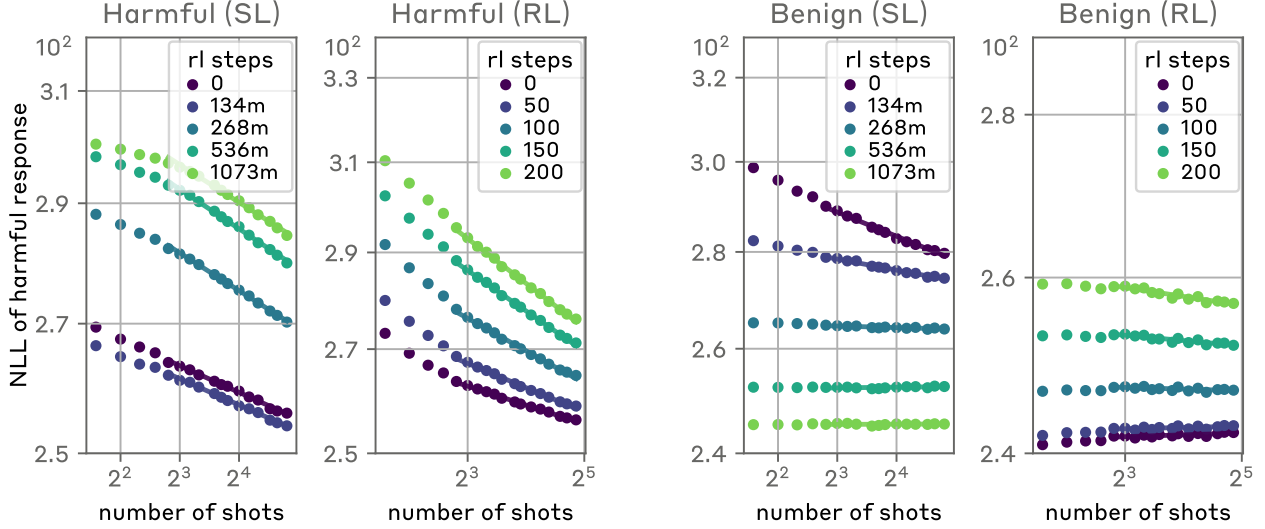


Figure 6. Supervised fine-tuning and reinforcement learning on examples that contain instances of MSJ only change the intercept. We ran supervised fine-tuning (SL) and reinforcement learning (RL) on a dataset that includes harmless responses to MSJ prompts. We then evaluated on prompts constructed with harmful and benign question-answer pairs. **(left)** The likelihood of zero-shot harmful responses decrease over the course of SL and RL (intercept goes up), while this effect is countered by the many-shot conditioning (slope remains similarly high). **(right)** For both SL and RL, the network learns the distribution of benign answers and does not benefit from in-context learning on benign examples (i.e. the slope converges to 0). The increase in the negative log-likelihood of benign responses during RL is likely due to the distribution shift between the learned policy and our evaluation data.

log-probability of harmful responses to many-shot attacks. Further details on the dataset composition and size, as well as training details are presented in Appendix E.

We first consider the effects of supervised training with such a dataset on the power laws for standard MSJ attacks (Figure 6 and 11). These attacks are made up of in-context examples of harmful responses to harmful requests and harmful responses to benign requests. We see that as supervised fine-tuning progresses, the zero-shot probability of a harmful response decreases, and the power law intercept increases. However, the power law exponent is largely unaffected. This implies that supervised finetuning to mitigate MSJ attacks is ineffective against protecting against MSJ with arbitrarily large context lengths. In other words, supervised finetuning in this way does not prevent the model from learning harmful behaviors from in-context patterns.

To gain further understanding, we also measure how the probability of harmful responses changes through supervised fine-tuning, but in cases where the in-context prompt explicitly encourages *benign* responses (Figure 6). Note that this is similar to the supervised training set we constructed, which also encourages benign responses. Here, we find that, given sufficient supervised fine-tuning, providing in-context examples of the desired behavior does *not* increase the probability of the desired behavior.

5.3. Mitigating by targeted reinforcement learning

We now explore reinforcement learning (RL) as a potential remedy for MSJ. We use a similar setup as the general RL results in Section 5.1, training a model on a set of general human/assistant data via RLHF. However, here we replace the standard harmlessness portion in the prompt mix with MSJ prompts, up to 10 shots long. Since MSJ may work on the model pre-RL, the model will produce harmful responses that are penalized by the preference model during RL. This experiment was run on a pre-RL snapshot of Claude Instant.

Targeted RL with MSJ prompts shows similar results to targeted supervised fine-tuning (Figure 6 and 12). The intercept of the power law on harmful requests increases, while the exponent remains unaffected. That is, while targeted RL makes the model less susceptible to zero-shot attacks, increasing the number of shots has a predictable and monotonic increase in the log-likelihood of harmful responses.

Unlike what happens during SL, however, the intercept on responses to benign requests *increases* during RL (Figure 6). One cause of this difference is that the RL results here are not from training exclusively on MSJ prompts,³ but rather other general examples of helpful behavior. RL may be bringing the model off-distribution with respect to the

³We confirm robustness to proportion of the prompt mix dedicated to MSJ; swapping up to 50% of the prompt mix results in no qualitative difference in results.

benign responses used in evaluation.

Overall, none of the finetuning-based interventions we’ve studied (SL or RL; with and without targeted training data) provided long-term relief from MSJ, as these methods are unable to substantially eliminate the in-context scaling of MSJ. Our results do not reject the idea that qualitative changes to existing finetuning pipelines might prove more effective against MSJ. However, doing so without causing unintended regressions will likely be challenging. Effective solutions should either reduce the slope, or increase the offset term ⁴ K of in-context power laws on harmful tasks.

5.4. Prompt-Based Mitigations

Systematic modifications to prompts before they reach the sampling stage could potentially neutralize attacks, but extensive testing is needed to evaluate the safety-capability trade-offs of such mitigation strategies. However, before conducting such testing, it’s worth exploring if any prompt-based defense can effectively thwart MSJ in the first place.

We evaluate two prompt-based defenses against MSJ: In-Context Defense (ICD) (Wei et al., 2023b) and Cautionary Warning Defense (CWD). ICD prepends the incoming prompt with demonstrations of refusals to harmful questions, while CWD prepends *and* appends natural language warning texts to caution the assistant model against being jailbroken. Our results (Appendix I) show that ICD only slightly reduces the attack success rate (61% to 54%) on the deception category of the malicious use-cases dataset with a 205-shot MSJ prompt, whereas CWD lowers the effectiveness to 2%. This trend is similar with shorter MSJ strings as well. Future work should evaluate the safety-capability trade-offs of Cautionary Warning Defense.

6. Related Work

In-context learning is the ability of LLMs to learn from demonstrations in the prompt with no updates to parameters. In-context learning performance typically increases with the number of provided examples. Xiong et al. (2023) show that language modeling loss decreases as function of the number of preceding tokens, following a power law. Fort (2023) describe a scaling law describing vulnerability of language models to adversarial perturbations of residual stream activations: the maximum number of output tokens that an attacker can control is directly proportional to the number of dimensions of activation space they can perturb. In our work, we observe that in-context learning follows power laws on most tasks, reminiscent of scaling laws for pretraining (Kaplan et al., 2020; Hoffmann et al., 2022) or finetuning (Hernandez et al., 2021) performance. Delétang et al. (2024) studies a similar quantity (in-context compression rate) as

⁴We also don’t observe significant shifts in the power law offset term K (Equation 1) during SL and RL.

a function of context length, but doesn’t comment on the functional form of their empirical results.

Previous work explored few-shot jailbreaking in the short-context regime, referring to it as In-Context Attack (Wei et al., 2023b) or Few-Shot Hacking (Rao et al., 2023). We prefer the phrase *many-shot* when referring to the long-context version of this attack to distinguish it from the short-context connotation of the phrase *few-shot*. We study the long-context scalability of this form of jailbreak by identifying the scaling laws and using these to measure progress towards fixing this vulnerability. The distinction between the fixed vs. arbitrarily long context versions of this attack makes a material difference on mitigation attempts. Our analyses suggest that the current alignment pipeline involving supervised and reinforcement learning *is* sufficient at mitigating the short-context version of MSJ, and fails with long context windows. Kandpal et al. (2023) explore how to conduct backdoor attacks such that the planted backdoor still gets activated through in-context learning. The theoretical results of Wolf et al. (2023) are also relevant to our findings. They show that, under the assumption that LLMs do Bayesian inference over their context, there exists a prompt with sufficient length that can elicit any behavior the model is capable of. We survey other major categories of language model jailbreaks in Appendix J.

7. Independent Replication on HarmBench

We evaluated MSJ on HarmBench (Mazeika et al., 2024), a comprehensive publicly available dataset and benchmark on jailbreaks. This evaluation was conducted by an independent part of our team, on an independent codebase and with subtly different design choices involving how to execute the attack. In this sense, these results can be thought of as an unofficial attempt at replicating our findings.

Our HarmBench results can be found in Appendix K. The experiments are conducted on Claude 2.0 (through Anthropic’s API). The takeaways are consistent with the rest of the paper. We find that among all the jailbreaking techniques considered in HarmBench, MSJ has a higher attack success rate, sometimes by a wide margin. We also replicate our findings regarding the importance of prompt diversity and composition of MSJ with other jailbreaking techniques.

8. Conclusion

Long contexts represent a new front in the struggle to control LLMs. We explored a family of attacks that are newly feasible due to longer context lengths, as well as candidate mitigations. We found that the effectiveness of attacks, and of in-context learning more generally, could be characterized by simple power laws. This provides a richer source of feedback for mitigating long-context attacks than the standard approach of measuring frequency of success.

Author Contributions

Cem Anil conceived and led this project, including developing the idea, characterizing the scaling laws, running the core experiments for the evaluation, analysis and mitigation of Many-shot Jailbreaking and writing the paper. **Esin Durmus** developed the Malicious Use-cases Dataset that was used to establish the severity of MSJ, and ran the majority of evaluations on this dataset. **Mrinank Sharma** worked on MSJ mitigations and had major contributions on writing the paper. **Joe Benton** replicated the GCG attack (Zou et al., 2023), worked on the theoretical analysis of the in-context power laws and contributed to the writing of the paper. **Sandipan Kundu** developed the Bounded Power Law and ran the model size - number of shot equivalence experiments. **Joshua Batson** developed the toy model to study in-context power laws. **Nina Rimskey** is involved in ongoing efforts to find alternative mitigations to MSJ. **Meg Tong** helped designed and produced the plots in the paper. **Jesse Mu** ran the reinforcement learning-based mitigation experiments involving synthetic data and contributed to the writing of the paper. **Francesco Mosconi** developed and evaluated the Cautionary Warning Defense.

Rajashree Agrawal lead the HarmBench replication attempts and designed the experiments around measuring the importance of prompt diversity and composition with other attacks on HarmBench. **Rylan Schaeffer**, **Naomi Bashkansky** and **Samuel Svenningsen** ran the experiments involving OpenAI, Meta and Mistral’s models. **Timothy Maxwell** and **Nicholas Schiefer** built the infrastructure on which most of the core experiments were run. **Tomasz Korbak** helped with background research and wrote the majority of the related works section.

Jared Kaplan and **Samuel R. Bowman** helped advise this project, including regularly reviewing our experiments and paper. **Deep Ganguli** wrote the Broader Impacts statement.

Ethan Perez was critical in launching this project, including helping with the initial positive results and the initial replication of MSJ on non-Anthropropic models. He also provided guidance on directions and experiments. **David Duvenaud** and **Roger Grosse** oversaw this project, providing detailed guidance on directions, experiments and presentation. Roger also articulated the main scientific arguments and developed the theory of impact around this research direction, and David helped develop a mechanistic explanation for the in-context power laws and write the paper.

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Broader Impacts

Alignment methods are promising (but nascent) techniques for preventing harm from models; however, our results provide further evidence that these methods still have major shortcomings. Alignment failure in general has broad implications for the **research community**, **model developers**, **malicious actors**, and possibly **malicious models**.

For the **research community**, we anticipate that our disclosure and characterization of the effectiveness of MSJ will help in developing methods to mitigate harms from this type of attack. We hope our work inspires the community to develop a predictive theory for why MSJ works, followed by a theoretically justified and empirically validated mitigation strategy. It is also possible that MSJ cannot be fully mitigated. In this case, our findings could influence public policy to further and more strongly encourage responsible development and deployment of advanced AI systems.

For **model developers**, our work offers a cautionary tale: we show that a seemingly innocuous update to a production model (in our case, adding longer context length) can open attack surfaces unanticipated/unexplored by developers prior to deployment. Given this observation, a broader impact of our work is to encourage developers to adopt a healthy red-team blue-team dynamic. The blue-team attempts to ensure the safety of seemingly minor product updates, and the red-team attempts to discover novel exploits. Such a dynamic may help elucidate and address safety failures prior to deployment.

Furthermore, our work raises an important question around fine-tuning. Developers may have strong economic incentives to allow downstream users to fine-tune models for specific purposes. At the same time, it is increasingly apparent that fine-tuning may override safety training (Qi et al., 2023). Our work suggests that even simpler and cheaper in-context learning is sufficient to override safety training. We anticipate a broader impact of our work is a stronger call to action for model developers to more closely consider the safety challenges inherent to offering both long-context windows for in-context learning in addition to offering fine-tuning capabilities.

Our work identifies a new jailbreak method that **malicious actors** can easily adopt to overcome safety measures implemented on publicly available models. Although the harms from jailbroken models are significant (Bommasani et al., 2022; Bender et al., 2021; Weidinger et al., 2021), we believe the benefits of disclosing and characterizing new jailbreaks currently outweigh the risk that new jailbreak methods are widely adopted. We are at a peculiar moment in time where we as a society are learning together how to both use and mis-use novel large language models. Model deployment in high-stakes domains (e.g., defense, healthcare, civil infrastructure, etc.) is currently minimal, but is likely